



Exploring 3D Native Foundation Models with Limited 3D Data by Unifying Text to 2D and 3D Generation

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Abstract

Recent multimodal large language models have achieved remarkable success in unified text and image understanding and generation, yet extending such native capability to 3D remains an open challenge. The core bottleneck is data: compared to the near-infinite 2D imagery on the web, high-quality 3D assets are orders of magnitude scarcer, leaving 3D synthesis severely under-constrained. Current methods often circumvent this limitation through indirect pipelines that edit in 2D image space and lift results into 3D via iterative optimization, sacrificing geometric consistency and the straightforwardness of native generation. We present Omni123, a 3D native foundation model that addresses limited 3D data by unifying text-to-2D and text-to-3D generation within a single autoregressive framework. Our key insight is that cross-modal generative consistency between images and 3D can act as an implicit structural constraint: by representing text, images, and 3D geometry as discrete tokens in a shared sequence space, the model can leverage abundant 2D observations as a rich geometric prior to fortify 3D representations. To realize this, we introduce an interleaved X-to-X training paradigm that coordinates diverse cross-modal tasks over heterogeneous paired datasets without requiring fully aligned text-image-3D triplets. By traversing semantic-visual-geometric cycles (e.g., text $\rightarrow$ image $\rightarrow$ 3D $\rightarrow$ image) within single autoregressive sequences, the model learns representations that simultaneously satisfy high-level semantic intent, appearance fidelity, and multi-view geometric consistency, while mitigating harmful interference between appearance and geometry objectives. Extensive experiments demonstrate that Omni123 significantly improves geometric consistency and semantic alignment in text-guided 3D generation and editing, validating that unifying 2D and 3D generative processes provides an effective and scalable pathway toward multimodal 3D world models.

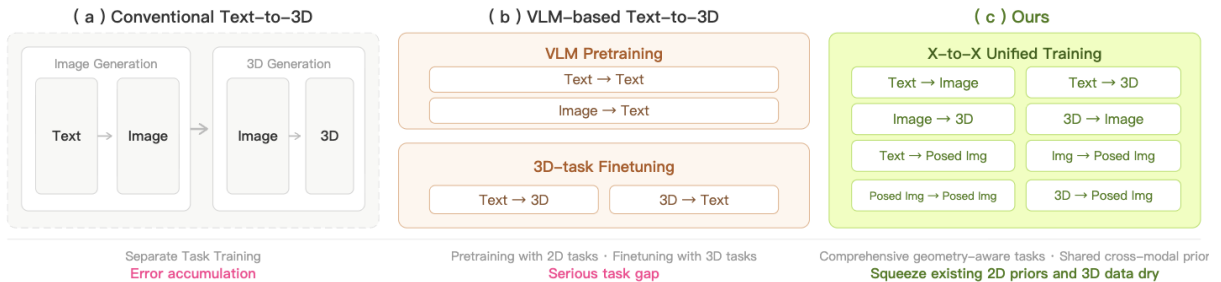
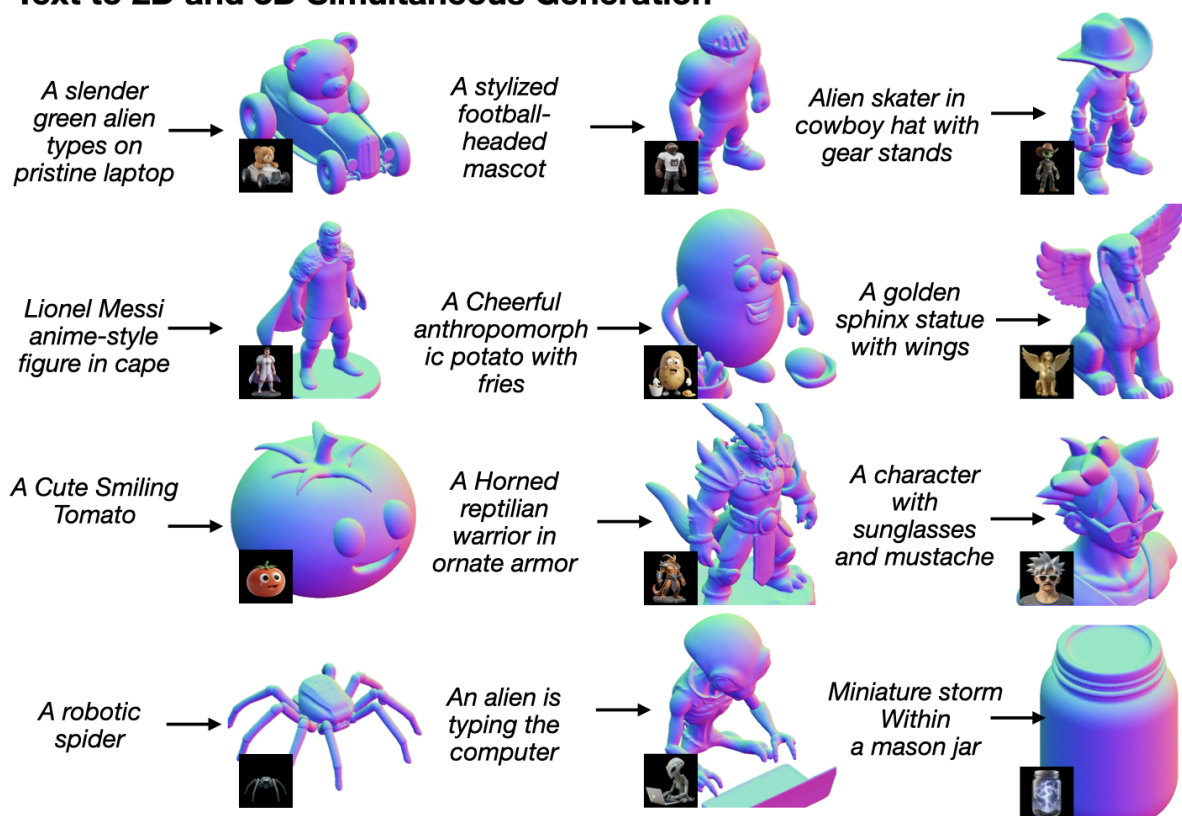


Figure 1 Comparison of Text-to-3D paradigms

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Text to 2D and 3D Simultaneous Generation



Instruction-Based Native 3D Editing

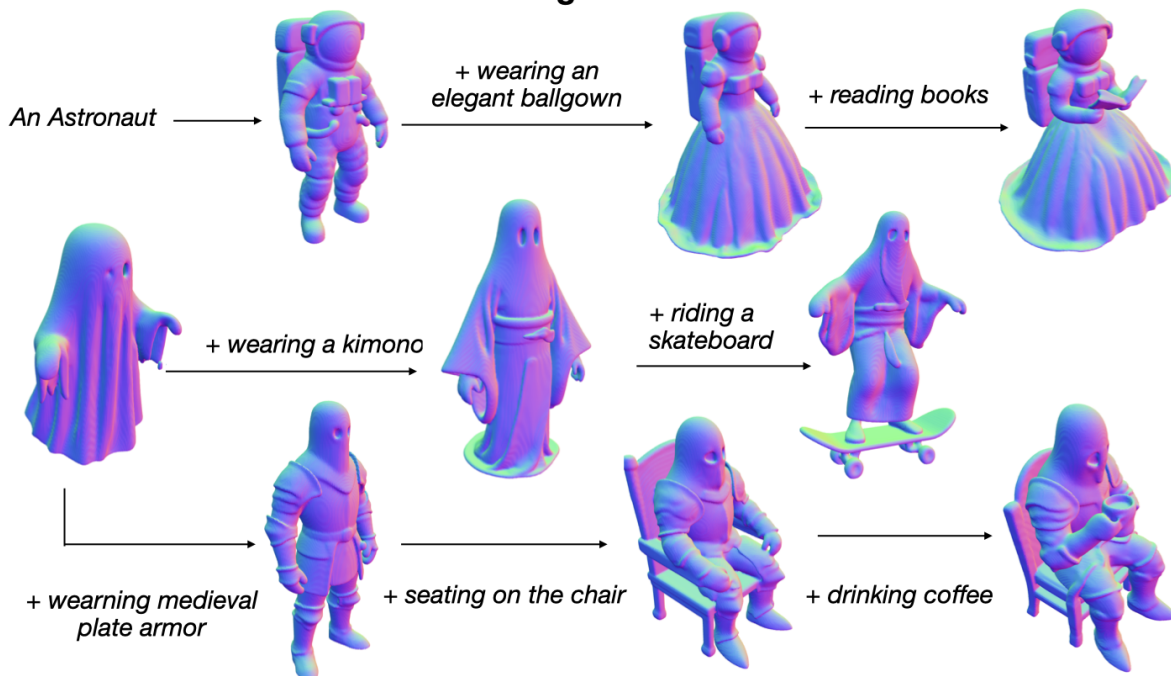


Figure 2 Omni123 enables native 3D generation and editing within a unified multimodal framework trained with limited 3D data. Top: Text-to-joint-2D-and-3D generation results across diverse prompts. Each example shows the generated 2D image (inset) alongside the normal map of the corresponding 3D output, demonstrating high geometric fidelity and strong semantic alignment with the input text. Bottom: Native 3D editing of 3D models through sequential and branching text instructions (e.g., “+wearing a kimono,” “+riding a skateboard”), illustrating the model’s ability to perform iterative and diverse 3D modifications directly in 3D space. More qualitative results are shown in Figures 8 and 9.

1 Introduction

Recent breakthroughs in generative AI have catalyzed a shift toward unified multimodal models capable of jointly reasoning across linguistic and visual domains, with models like GPT-4o[52] and Nano-Banana [23] enabling native capabilities for high-fidelity synthesis and instruction-guided editing. As the community pushes toward more immersive digital environments, the natural evolution is the quest for a "3D Nano-Banana": a unified framework capable of native 3D generation and editing, which is essential for advancing applications in embodied AI and autonomous virtual world synthesis. However, achieving such native capability in 3D remains a persistent challenge.

Unlike 2D AIGC, which leverages near-infinite web-scale data, 3D generation faces acute data scarcity. This disparity leaves 3D synthesis inherently under-constrained, especially under sparse supervision from text or single views. To compensate, current methods often treat 2D images as an intermediate proxy—performing edits in the image space and lifting them into 3D through iterative optimization. While effective, these indirect approaches frequently compromise geometric consistency and lack the straightforwardness of 2D workflows. This suggests that 3D learning requires stronger structural constraints than sparse signals can provide. Crucially, as 2D visual data implicitly encodes deep 3D structural information, including object shape [43], appearance [73], and spatial relationships [3], it prompts a pivotal hypothesis: Can we leverage the abundance of 2D observations to fortify 3D generation via a unified multimodal framework?

A common strategy involves aggregating multiple cross-modal tasks. However, it does not inherently guarantee effective knowledge transfer. This stems from the fact that different generative tasks are rooted in distinct statistical prior: text-to-image task focuses on modeling appearance distributions, whereas text-to-3D task requires consistent geometric reasoning. Without meticulous coordination, joint training may introduce both **beneficial transfer and harmful task interference**, where competing gradients may ultimately degrade the quality of the learned 3D representations.

To address this challenge, we introduce an **interleaved X-to-X training paradigm**. We **posit that cross-modal generative consistency between images and 3D can act as an implicit structural constraint for learning robust 3D representations**. Different modalities offer complementary observations of the same underlying object: text describes high-level semantic intent, images capture rich appearance cues and partial geometric hints, while 3D representations provide explicit spatial structure and multi-view consistency. Consequently, we propose that cross-modal generative consistency between images and 3D can serve as a potent implicit constraint. Specifically, by forcing a model to traverse a "semantic-visual-geometric" cycle (e.g., text $\rightarrow$ image $\rightarrow$ 3D $\rightarrow$ image), the resulting representations could simultaneously satisfy high-level intent, appearance fidelity, and multi-view consistency.

We investigate this hypothesis through a unified autoregressive formulation that treats text, image, and 3D geometry as discrete tokens within a shared sequence space. To circumvent the reliance on fully paired text-image-3D triplets, we leverages heterogeneous paired datasets. The model is trained on a diverse set of conditional generation tasks where any modality can serve as input and the target modality is predicted autoregressively. This formulation maximizes the utility of available multimodal data and ensures that the model does not merely learn isolated tasks, but rather develops a unified latent representations across the semantic, visual, and geometric domains.

In summary, we incorporate the following key technical advancements:

- We introduce a framework that tokenizes text, images, and 3D geometry within a shared sequence space. This enables native 3D reasoning and editing, eliminating the reliance on intermediate 2D proxies and iterative image-space optimization.
- We propose an interleaved training scheme that coordinates diverse generative tasks across

heterogeneous datasets. This approach mitigates task interference between appearance and geometry priors, facilitating beneficial cross-modal knowledge transfer from 2D to 3D.

- We leverage generative consistency across "semantic-visual-geometric" cycles as an implicit structural constraint. This allows abundant 2D observations to serve as a rich geometric prior, effectively fortifying 3D representations against acute data scarcity.

Through extensive experiments, we study how different multimodal objectives interact during training and how cross-modal generative processes influence the quality of learned 3D representations. Our results indicate that interleaved cross-modal generation significantly improves geometric consistency and semantic alignment in text-guided 3D generation and editing tasks. These findings suggest that **cross-modal generative consistency provides an effective pathway toward scalable multimodal 3D world models.**

2 Related Work

2.1 Image Generation & Editing

Generative adversarial network (GAN) is the first representative image generation [5, 21, 34, 35] and editing [44, 53] method in deep learning era. To improve the distribution coverage, subsequent research also explored likelihood-based generative models. Diffusion models [8, 17, 24, 54, 61] and Flow Matching [19, 37, 45] later became the dominant paradigm by formulating image synthesis as the reverse denoising process from noise to data. Beyond their strong generation quality, diffusion models provide a more expressive latent space and a natural interface for conditioning, which has driven the rapid development of image editing methods, including controllable generation[68, 93], inversion-based training-free editing[12, 20], and optimization-based editing[32, 60]. More recently, emerging large-scale visual generation models[7, 38, 76] have increasingly explored leveraging stronger multi-modal reasoning and instruction-following capabilities of VLMs to unify image editing, generation, and understanding.

2.2 3D Generation & Editing

3D generation has evolved along two main paradigms. [46, 57] relies on pretrained image/video diffusion models and distills their priors into 3D representations through score distillation. While effective, this indirect 2D supervision often leads to slow optimization, inconsistent 3D appearance and artifacts. In contrast, recent native 3D generative models [27, 78, 79] learn 3D distribution directly, providing a more efficient and scalable route toward high-quality 3D synthesis and already seeing broad adoption in commercial applications[30, 40, 51, 71]. Building on these strong foundation models, many recent works [26, 28, 89] have further explored editing tasks. Nevertheless, direct end-to-end text-3D synthesis [25, 55, 58] still lags far behind due to the large modality gap. To address this challenge, we propose a text-image-3D-image interleaved training paradigm that uses images as a bridge to alleviate both the modality gap and the conflict between generation and reconstruction, achieving state-of-the-art text-3D performance.

2.3 Multimodal Generation

Numerous previous works have explored extending LLMs toward visual generation and understanding via mapping learned visual tokens into the text feature space of a pretrained diffusion [11, 36, 67]. Later studies further explore an end-to-end unified architecture for generation and understanding across modalities [9, 15, 47, 74, 83]. More recently GPT-4o [52] and Nano-Banana [23] have further accelerated this progress.

For the 3D modality, ShapeLLM-Omni [88] is among the first to unify 3D generation and understanding within a single autoregressive framework. However, it purely suffers from quantization-induced information loss, which limits generation quality. Lucid3D [10] further extends to 3D editing, while Cube3D[4] improves the 3D tokenizer with self-supervised objectives, leading to higher fidelity and further extending to 3D scene generation. AR3D-R1[69] introduces the first 3D RL system to improve 3D consistency and alignment with human preferences. However, due to the scarcity and expense of 3D data, the scalability and generalization of previous works have been constrained. To alleviate this limitation, we unify 2D and 3D generation so that 3D tasks can benefit from rich and general visual knowledge from large-scale 2D data. This enables high-quality 3D creation with only limited text-image-3D pairs, while also yielding emergent generation and editing capabilities.

3 Data Pre-Processing

In this section, we describe the composition of our multi-modal dataset, the cleaning and filtering pipelines applied to each modality, the data synthesis pipeline used to scale high-quality SFT data, and the captioning strategies used to bridge modalities.

3.1 Data Overview

Our training corpus spans three modalities—text, images, and 3D assets—and is organized into the following paired and triplet subsets. Table 1 summarizes the complete data composition across pre-training and supervised fine-tuning stages.

Table 1 Summary of training data composition. Token counts per sample are computed using 77 text tokens, 1,024 image tokens per view, and 1,024 3D mesh tokens. The table is divided into pairwise pre-training data (upper) and continued training & supervised fine-tuning data (lower). Total tokens shown are the dataset size; tokens seen during training are higher due to multiple epochs (see Table 6).

Stage	Data Configuration	Items	Tokens/Sample	Total Tokens	Token %
Pre-Training	Text + Image	63.7M	1,101	70.1B	14.8%
	Image + 3D	120M	2,048	245.8B	51.8%
	Text + 3D	9M	1,101	9.9B	2.1%
	<i>Subtotal</i>	<i>192.7M</i>	—	<i>325.8B</i>	<i>68.6%</i>
CT & SFT	Text + Single-View Image + 3D	2M	2,125	4.3B	0.9%
	Text + 6 Posed Images + 3D	20M	7,245	144.9B	30.5%
	<i>Subtotal</i>	<i>22M</i>	—	<i>149.2B</i>	<i>31.4%</i>
Total		214.7M	—	475.0B	100%

The pre-training stage uses 194.7 M pairwise samples spanning three cross-modal directions (Text $\leftrightarrow$ Image, Image $\leftrightarrow$ 3D, Text $\leftrightarrow$ 3D), totaling 328.0 B tokens in the dataset itself. The interleaved SFT stage introduces two chain-of-generation tasks: (i) a single-view pipeline (Text $\rightarrow$ Image $\rightarrow$ 3D) with 2 M triplets, and (ii) a multi-view pipeline (Text $\rightarrow$ 6 Posed Images $\rightarrow$ 3D) with 20 M samples, where six views are rendered from known camera poses. Despite comprising only 9.2% of all items, the multi-view SFT data accounts for 30.4% of all tokens due to the much longer per-sample sequence length (7,245 tokens). Note that the total tokens seen during training (reported in Table 6) is higher (~ 1.16 T for pre-training) due to multiple epochs over the data. Table 2 details the tokenization configuration and sequence lengths for each modality.

3.2 Text–Image Pairs

The text–image pre-training corpus contains 63.7 M open-domain images in total, comprising proprietary data mixed with synthetic data by Z-Image [7].

A natural question is whether the 120 M rendered images from our Image–3D corpus (Section 3.3) should also be included, given their scale. We deliberately choose *not* to do so: these renderings exhibit uniform lighting, synthetic material appearance, and the absence of natural backgrounds, diverging substantially from in-the-wild photographs. Mixing the two domains in the text–image objective would create a distribution conflict that degrades image generation quality. We therefore confine rendered imagery to the Image $\rightarrow$ 3D and Text $\rightarrow$ 3D objectives, where the rendering distribution is consistent with the target task.

3.3 Image–3D Pairs

The image–3D pairs constitute the largest subset (120 M) and are critical for learning the reconstruction mapping between 2D observations and 3D geometry. Following established

Table 2 Per-modality tokenization configuration.

Modality	Tokens	Tokenizer
Text	77	CLIP [59] text encoder + Qwen3-0.6B [86] for enhanced semantic understanding
Image	1,024 / view	Our image tokenizer, a 1D discrete visual tokenizer with semantic regularization
3D Mesh	1,024	Cube3D [4], a Perceiver-based VQ-VAE with optimal-transport vector quantization

practices in recent large-scale 3D data engineering [29, 41, 42, 92], we process this corpus through a three-stage pipeline:

(1) *Rendering and format conversion*: 3D assets from heterogeneous sources are converted into meshes, aligned to a canonical orientation, standardized with PBR textures, saved as GLB files, and rendered using Meshy’s in-house renderer with HDRI environments and lighting conditions sampled from a pool of 2,000 setups.

(2) *Asset filtering*: we exclude assets with poor geometric quality (low polygon counts, simplistic topology), poor texture quality (UV defects, low resolution), noisy photogrammetry scans, and large thin-walled structures—the latter causing SDF sign discontinuities that destabilize training and multi-view inconsistencies where thin geometry becomes invisible under certain viewpoints.

(3) *Post-processing*: filtered meshes undergo watertight conversion, narrow-band SDF sampling, and point cloud sampling.

3.4 Text–3D Pairs

Existing captions for 3D objects [49] either suffer from poor alignment with the geometry they describe or lack sufficient detail, which limits high-quality text-to-3D generation. We therefore design a multi-granularity captioning pipeline utilizing visual chain-of-thought reasoning, to produce multiple precise and detailed text descriptions focusing on different aspects of a 3D model. When applicable, metadata is also supplied to the model for factual world knowledge grounding.

The pipeline is run on each of the 5 M filtered 3D assets, consisting of three stages:

(1) *Visual Chain of Thought Analysis*: we feed multi-view rendered images of each asset into a Visual-Language-Model (VLM) such as Gemini-3 [22] to generate highly detailed per-view descriptions that are used as anchors for the model to perform orientation and spatial correspondence analysis. After the model has identified spatial correspondence between images and therefore oriented the object, the VLM is asked to analyze the model in terms of its appearance, geometry, potential function and origin in the real world.

(2) *Captioning and Categorization*: The VLM takes the analysis and produces a paragraph-long detailed caption that goes over all aspects of the 3D model. Categories are then assigned by the VLM producing rough semantic classification.

(3) *Human captions mimicking*: We prepared 10K human labeled short captions distributed over all categories, and 4 random examples are selected per-asset based on its evaluated category to perform few-shot caption augmentation. Human captions leveraged in this way greatly improve the lexical and syntactical diversity of our captions.

All three stages are completed within one continuous multi-turn conversation, preserving all context to the model — including the images. Reasoning is enabled, and the reasoning framework is enforced through schema-guided generation.

Through this multi-granularity captioning process, each of the 5 M filtered 3D assets is augmented with multiple caption variants. We then perform an alignment filtering step, removing pairs where the caption poorly matches the 3D geometry based on VLM-evaluated consistency scores. This yields a total of 9 M Text–3D training pairs used in pre-training (Section 4.4).

3.5 Interleaved SFT Triplets

While the Objaverse corpus [13, 14] provides a substantial volume of image–3D training pairs, the majority of its assets exhibit simple structures and plain surface details, limiting the generation capabilities of models trained exclusively on this data. Given the prohibitive cost of manually creating high-quality 3D assets, we construct a data synthesis pipeline that performs Text→Image→3D generation, producing a dataset of synthesized 3D assets with considerably more complex structures and richer geometric detail.

Step 1: Prompt Curation. We initiate the synthesis process from text prompts rather than image prompts, as text affords more direct control over semantic diversity and ensures variety in the resulting geometries. We first source approximately 14 M high-quality raw prompts from DiffusionDB [75]. A Gemini-3-Flash [22] classifier is applied to filter out complex multi-object scenes, retaining only single-object descriptions amenable to 3D asset generation. Rule-based filtering further eliminates stylistic modifiers (e.g., “cinematic”, “4k wallpaper”), followed by a structural standardization step to enforce consistent prompt formatting. After all filtering stages, approximately 2 M prompts are retained.

Step 2: Text-to-Image Generation. We employ Z-Image-Turbo[7] for high-throughput text-to-image generation. The text prompt conditioning is augmented with viewpoint and lighting specifications to produce object-centric images with controlled camera angles, which is important for stable downstream 3D reconstruction.

Step 3: Background Removal. To obtain clean object masks suitable for image-to-3D generation, we apply BiRefNet [95] for high-resolution foreground segmentation. We then subsequently use VLM to evaluate each masked image for segmentation artifacts (e.g., residual background, truncated object parts) and filters out samples with poor mask quality. It also simultaneously re-captions the masked object to produce an aligned text description, ensuring cross-modal consistency between the final caption and the visual content.

Step 4: Image-to-3D Synthesis. We adopt Meshy 4 to synthesize 3D meshes from the masked images. The generated meshes undergo the same watertight processing and point cloud sampling pipeline described in Section 3.3.

Step 5: Quality Filtering. A rigorous data cleaning process combines automated geometric assessment with MLLM-based evaluation. Specifically, we render multi-view images of each synthesized mesh and evaluate three quality axes: (i) *geometric consistency*—whether the object is structurally sound with preserved symmetry and no warped geometry; (ii) *surface texture clarity*—whether micro-details (e.g., wood grain, metal finish, fabric weave) are sharp rather than flat or muddy; and (iii) *silhouette connectivity*—whether the object has a clear, recognizable silhouette without heavy occlusion or detached floaters. For each axis, the MLLM assigns a binary “high”/“low” rating, and only samples rated “high” on all three axes are retained in the final dataset.

Multi-Granularity Re-Captioning. Each retained asset is re-captioned at three levels of specificity by an MLLM: a *long* caption (40–70 words) providing specification-level detail including materials, textures, colors, and geometric complexity; a *medium* caption (10–25 words) covering subject, primary material, and basic geometry; and a *short* caption (1–5 words) capturing only the core subject identity. These multi-granularity captions serve both the SFT triplet task (Text→Image→3D) and the multi-view SFT task (Text→6 Posed Images→3D).

4 Model Training

4.1 Architecture

A key obstacle to unified multi-modal generation is the heterogeneity of representations across modalities. Text is naturally a sequence of discrete symbols; images are typically encoded as 2D spatial grids of continuous features [1, 61]; and 3D shapes are represented as volumetric lattices [88], unordered point sets, or continuous latent vectors [91]. These representations are incompatible with autoregressive language modeling in two fundamental ways. First, *continuous* latent spaces—such as the latent vectors produced by 3DShape2VecSet [91]—lack the discrete vocabulary required for next-token prediction; Second, even when discretized, *2D or 3D spatial* tokenizers produce token counts that scale with spatial resolution, yielding intractable context lengths, and their modality-specific positional encodings and attention patterns fragment the model and hinder cross-modal knowledge transfer.

Building on recent advances in *vector-quantized* tokenization [4, 84] that convert continuous representations into discrete codebook indices amenable to next-token prediction, and *1D* tokenization that compresses each modality into a short, structurally uniform sequence, Omni123 adopts a self-developed representative image tokenizer (Section 4.3) for images and Cube3D [4] geometry tokenizer for 3D meshes. This combination yields compact 1D discrete token sequences for both modalities, enabling seamless concatenation without modality-specific positional encoding. Both modalities are then processed by a shared autoregressive Transformer backbone that maximizes parameter sharing and enforces cross-modal coherence through shared self-attention.

As illustrated in Figure 3, the architecture consists of modality-specific tokenizers, a unified autoregressive Transformer backbone, and modality-specific output heads. For text conditioning, we employ two complementary encoders: CLIP [59] provides $L_{\text{clip}} = 77$ vision-language-aligned embeddings, while Qwen3-0.6B [86] provides L_{qwen} embeddings that capture richer linguistic detail; the two are concatenated and injected via cross-attention. Images are tokenized into $L_i = 1,024$ 1D tokens via our image tokenizer; and 3D geometry are encoded into $L_m = 1,024$ 1D *geometry tokens* via Cube3D [4] shape tokenizer (detailed in Sections 4.2 and 4.3). Image and 3D shape tokens are concatenated at the sequence level and processed by the backbone autoregressively.

The backbone follows the dual-stream autoregressive architecture of Cube3D [4]. A *conditioning stream* carrying text embeddings and a *generation stream* carrying image and 3D shape tokens run in parallel through the dual-stream layers. At each dual-stream layer, the two streams are independently projected into queries, keys, and values; these are then concatenated along the sequence dimension and processed by a single joint attention operation under a causal mask. The attention output is split back along the sequence boundary and used to update both streams through separate feed-forward sub-layers. In the final dual-stream layer, the conditioning stream contributes only keys and values—its query projection and post-attention update are omitted, so the information flow becomes unidirectional from conditioning to generation. The subsequent single-stream layers retain only causal self-attention over the generation tokens, without further text interaction. Within the generation stream, image and 3D shape tokens are concatenated into a single flat sequence and processed by fully shared self-attention weights—no modality-specific branches or attention masks are introduced. This means every attention layer implicitly performs cross-modal fusion, and all generation-side parameters are shared across modalities, enabling visual priors learned from the large-scale Text→Image data to directly benefit the data-scarcer Text→3D task. Each transformer layer employs SwiGLU [62] as the feed-forward network with a $4\times$ expansion ratio, following the design of Cube3D [?]. Detailed architecture hyperparameters are listed in Table 3.

Table 3 Model architecture configuration.

Hyperparameter	Value
Dual-stream (joint-attention) layers	24
Single-stream (self-attention) layers	6
Hidden dimension d	1,536
Attention heads	12
Per-head dimension	128
FFN expansion ratio	$4\times$ (SwiGLU)
Text encoder (vision-language)	CLIP ($L_{\text{clip}} = 77$)
Text encoder (linguistic)	Qwen3-0.6B
Image tokens L_i	1,024
3D shape tokens L_m	1,024
3D codebook embedding dim	16
RoPE θ	1,000
Layer norm type	LayerNorm ($\epsilon = 10^{-6}$, no affine)
QK-Norm	RMSNorm
Token dropout p_{drop}	0.1

4.2 3D Shape Tokenizer

A central requirement of our unified architecture is a 3D tokenizer that produces discrete tokens capable of faithfully capturing a wide range of geometric properties—smooth surfaces, sharp edges, high-frequency details—while being natively compatible with autoregressive sequence models. We adopt Cube3D [4], a shape tokenizer built on the 3DShape2VecSet [91] framework, which converts continuous shape representations into discrete tokens suitable for mixed-modal foundation models [70].

Encoder-Decoder Architecture. The tokenizer follows an encoder-decoder design. The encoder takes as input a point cloud sampled from the mesh surface and embeds it using a phase-modulated positional encoding, which improves the Perceiver-based transformer’s ability to disambiguate spatially distinct points in cross-attention layers. A Perceiver-based transformer [31] compresses the point cloud into a set of $L_m = 1,024$ continuous latent vectors, which are subsequently discretized through optimal-transport vector quantization [4] to produce discrete 3D shape tokens. The decoder reconstructs an implicit occupancy field from these discrete tokens, from which a triangle mesh can be extracted via Marching Cubes.

4.3 Image Tokenizer

Image tokenizers [63, 72] play a crucial role in compressing visual data into compact discrete latent tokens, facilitating scalable and efficient training of autoregressive models for visual generation via next-token prediction. We build our image tokenizer upon GiGaTok [84], whose 1D token representation is more compatible with our 3D shape tokenizer, while also better handling the inherent redundancy in images [90]. Recent studies [97] trace the sub-optimal performance of discrete tokenizers back to a core limitation: the disjoint optimization of the codebook. This disjointness inherently degrades reconstruction fidelity compared to continuous VAEs [61, 87], ultimately acting as a critical bottleneck that hinders pure autoregressive models from matching the state-of-the-art generation quality of diffusion models or hybrid approaches. To alleviate this issue, we propose a two-stage training strategy (Figure 4): First, we train a continuous VAE to learn semantically rich visual representations and high-fidelity reconstruction. Then we insert the 1D GiGaTok [84] into the pretrained VAE and train it only to reconstruct the continuous features, simplifying the vector quantization to a pure 1D compact token extraction task. This

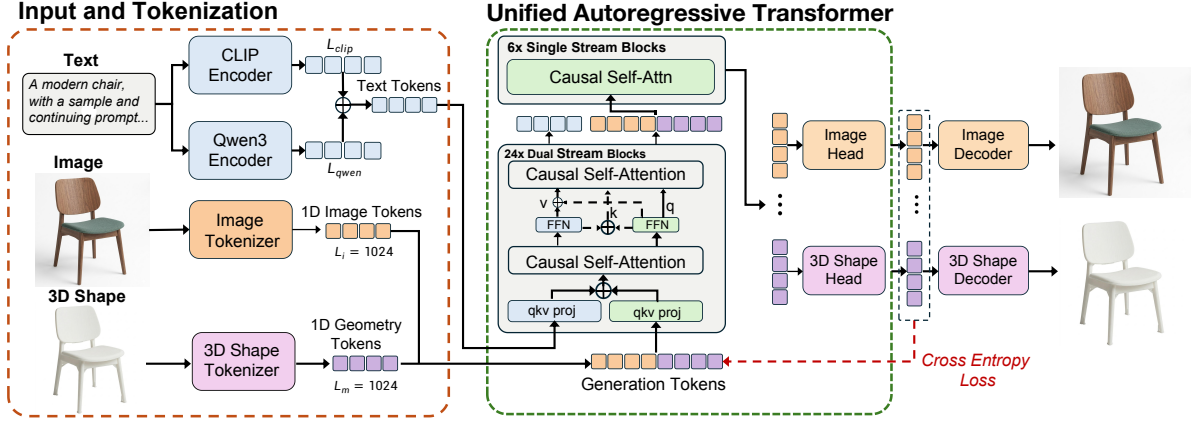


Figure 3 Overview of the **Omni123** architecture. Text is encoded by dual text encoders (CLIP [59] and Qwen3-0.6B [86]) and fed into a conditioning stream, while images and 3D shapes are tokenized into 1D discrete tokens and concatenated into a unified generation stream. The unified autoregressive transformer backbone uses 24 dual-stream blocks to jointly process the conditioning and generation tokens under causal attention, followed by 6 single-stream layers operating only on generation tokens, finally with modality-specific linear heads decoding token logits over the 2D and 3D codebooks.

design yields a high-performance tokenizer that is scalable and equipped with rich semantic representations.

Architectural Details. Inspired by recent successful Representation Autoencoder[18, 64, 94], we retrain DINO-Tok [33] with Dinov3-L [65] encoder and an 832-dimensional latent space as aforementioned VAE on ImageNet [16] with 256×256 and 512×512 resolution. In the second stage, we freeze the DINO-Tok VAE and only train the 1D Q-Former on a 60M-image dataset and 512×512 resolution with 1024 query tokens and a 64K-entry codebook of dimension 64. To the best of our knowledge, our tokenizer is the first 1D visual tokenizer trained at this scale.

4.4 Model Pre-training (PT)

The central challenge motivating our pre-training design is the acute scarcity of 3D data relative to 2D: our corpus contains 63.7M text–image pairs but only 9M text–3D pairs. As argued in Section 1, 2D observations implicitly encode rich 3D structural information—shape, appearance, and spatial relationships—suggesting that large-scale 2D supervision can serve as a powerful geometric prior for 3D learning. However, naïvely mixing appearance-focused and geometry-focused objectives risks task interference rather than beneficial transfer.

Our pre-training strategy addresses this tension through the *X-to-X* paradigm: all cross-modal generation tasks—text-to-image, text-to-3D, image-to-3D, and 3D-to-image—are unified as autoregressive next-token prediction over a shared discrete vocabulary. By training a single model to traverse multiple generation paths across the “semantic–visual–geometric” cycle, the framework enforces cross-modal generative consistency as an implicit structural constraint. Concretely, the large-scale Text→Image subset strengthens the shared text conditioning pathway that Text→3D relies on despite its limited data, while the 3D→Image task—as the inverse of Image→3D—forces the model to maintain explicit 3D-to-2D geometric correspondences, which in turn benefits forward Image→3D reconstruction.

Task Formulation. During pre-training, we train on four core generation tasks (Table 4). Each task takes one or two source modalities as conditioning and autoregressively generates tokens in the target modality. Text, when present, is injected via the cross-attention pathway; non-text source tokens are placed as a non-causal prefix in the self-attention sequence. The model is trained to predict only the target tokens that follow the prefix. Note that text conditioning

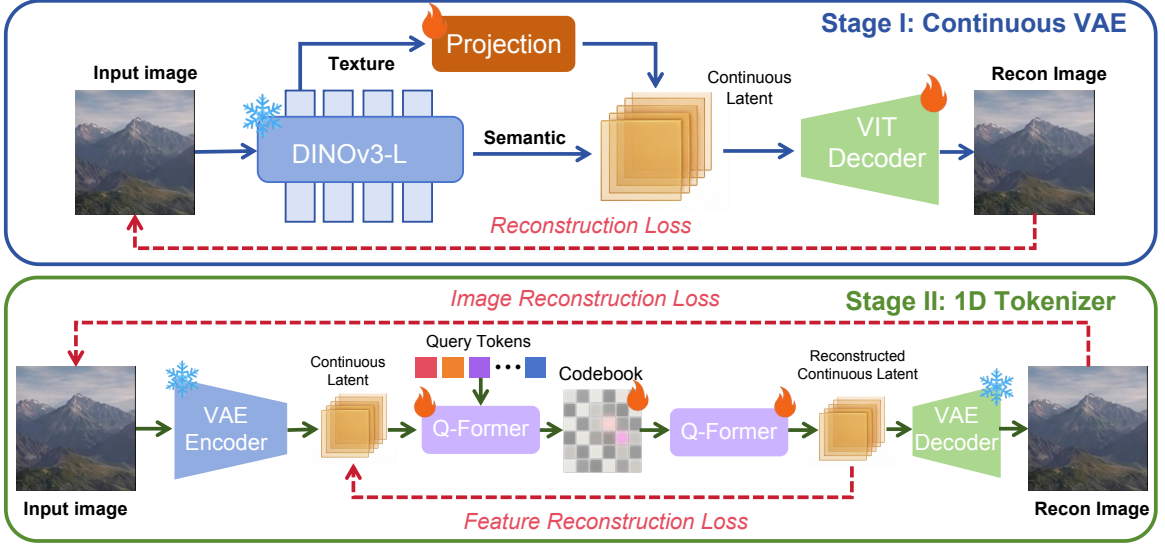


Figure 4 Two-stage image tokenizer training strategy. Stage 1 trains a continuous VAE (DINO-Tok) to learn high-fidelity visual representations. Stage 2 freezes the VAE and trains a 1D Q-Former to reconstruct the continuous features, reducing vector quantization to a compact 1D token extraction task.

Table 4 Pre-training task configurations. Text conditioning is optionally dropped for Classifier-Free Guidance. The loss is computed only over the target column.

Task	Cond. Stream	Prefix	Target
Text $\rightarrow$ Image	Caption	—	Image tokens
Text $\rightarrow$ 3D	Caption	—	3D shape tokens
(Text +) Image $\rightarrow$ 3D	(Caption)	Image tokens	3D shape tokens
(Text +) 3D $\rightarrow$ Image	(Caption)	3D shape tokens	Image tokens

is randomly dropped during training to enable Classifier-Free Guidance (discussed in Training Objective below), so the model must also learn to generate from the non-text prefix alone.

Training Objective. All four tasks share a single autoregressive cross-entropy loss over the target token sequence:

$$\mathcal{L} = - \sum_{j=1}^{L_{\text{tgt}}} \log p_{\theta}(x_j | x_{<j}, \mathbf{c}) \quad (1)$$

where x_j is the j -th target token, $x_{<j}$ denotes preceding tokens including any source-modality prefix, and $\mathbf{c}$ is the text conditioning (empty when dropped). A uniform loss weight is applied across all tasks. To enable Classifier-Free Guidance [?] at inference time, we randomly drop the text conditioning with probability $p_{\text{drop}} = 0.1$ during training [?], replacing it with a learned null embedding. For the two prefix-conditioned tasks (Image $\rightarrow$ 3D and 3D $\rightarrow$ Image), this means the model is exposed to three conditioning regimes: text + prefix, prefix only, and unconditional, allowing flexible guidance at inference.

Data Mixture. Pre-training draws from three data pools of varying scale: Text–Image pairs (63.7 M), Image–3D pairs (120 M), and Text–3D pairs (9 M). Each pool serves one or more tasks: Text–Image supports Text $\rightarrow$ Image; Image–3D supports both Image $\rightarrow$ 3D and 3D $\rightarrow$ Image (with the task direction randomly selected per sample); and Text–3D supports Text $\rightarrow$ 3D. Naïvely sampling in proportion to dataset size would severely under-represent the smallest yet critical Text–3D subset. We adopt a temperature-based weighted sampling strategy. The probability of

Table 5 Pre-training data mixture. Each pool may serve multiple tasks as described in the text.

Data Pool	Size	Tasks Served	Priority	Effective P_i
Text–Image	63.7 M	Text→Image	1.0	0.30
Image–3D	120 M	Image→3D, 3D→Image	1.5	0.53
Text–3D	9 M	Text→3D	3.0	0.17

Table 6 Training recipe across all stages.

Hyperparameter	Pre-training		Post-training	
	Stage 1	Stage 2	CT	SFT
Peak learning rate	5e-4	1e-4	1e-5	1e-6
LR scheduler	Cosine			
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)			
Weight decay	0.1			
Gradient norm clip	5.0			
Warm-up steps	5,000	1,000	1,000	2,000
Training steps	400,000	250,000	120,000	115,000
Effective batch size	2,048	1,024	1,024	256
Image resolution	256 ²	512 ²	512 ²	512 ²
GPUs	256×H100			
Tokens seen	~0.76 T	~0.40 T	~0.20 T	~0.15 T

drawing from subset i is:

$$P_i = \frac{(\text{Size}_i)^T \times \text{Priority}_i}{\sum_{j=1}^n ((\text{Size}_j)^T \times \text{Priority}_j)} \quad (2)$$

where Size_i is the number of samples in pool i , Priority_i is a manually assigned importance weight reflecting task value, and $T \in (0, 1]$ is a temperature controlling the degree of rebalancing—lower T flattens the size distribution, giving smaller but high-priority pools greater representation. Table 5 lists the configuration.

Training Schedule. Pre-training proceeds in two stages (Table 6). Stage 1 trains at 256×256 resolution for 400,000 steps with a peak learning rate of 5e-4, allowing the model to learn cross-modal alignment across all four tasks. Stage 2 increases the image resolution to 512×512 and continues for 250,000 steps with a reduced learning rate of 1e-4, refining visual fidelity while preserving the learned cross-modal representations. Both stages use AdamW [48] ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay = 0.1) with a cosine learning rate schedule. Gradient norms are clipped at 5.0. The effective batch size is 2,048 (Stage 1) and 1,024 (Stage 2) across 256×H100 GPUs. In total, the model sees approximately 1.16 T tokens during pre-training.

4.5 Continued Training (CT)

Pre-training establishes cross-modal alignment between text, image, and 3D representations, yet the model lacks an explicit notion of *viewpoint*: it can generate images and 3D shapes, but cannot reason about how the same object appears from different camera poses. This is a critical gap—multi-view consistency is the bridge that connects 2D appearance to 3D geometry, and without it the “semantic–visual–geometric” cycle introduced in Section 1 remains incomplete. Continued training (CT) closes this gap by introducing *view tokens* (Figure 5): a set of N learnable embeddings, each bound to a fixed canonical viewpoint (front, back, left, right, top, bottom). By prepending a view token to the target image sequence, the model learns to associate

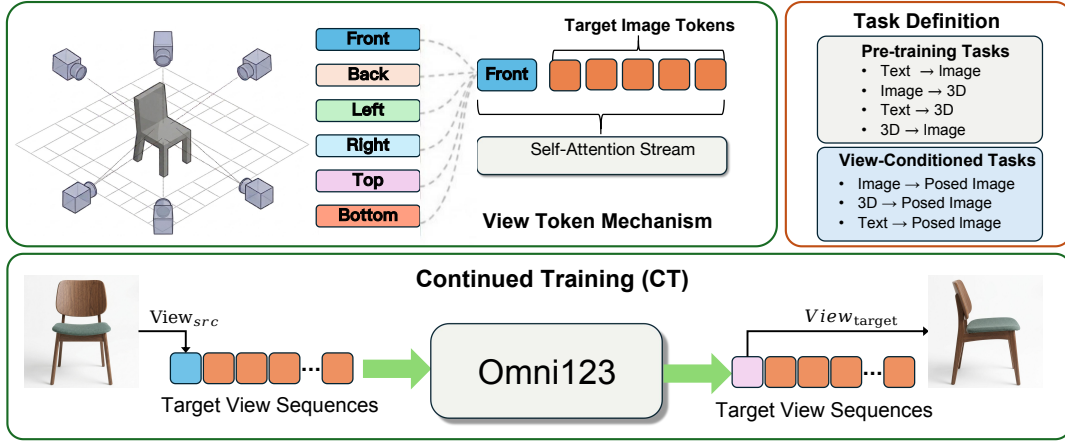


Figure 5 Continued training introduces view-conditioned generation via learnable view tokens. **Top left:** $N=6$ view tokens corresponding to canonical viewpoints (front, back, left, right, top, bottom) are mapped from 3D camera extrinsics. **Top right:** A view token $\mathbf{v}$ is prepended to the target image tokens and fed into the self-attention stream, enabling viewpoint-controllable generation. **Bottom:** In the Posed Image $\rightarrow$ Posed Image task, the source image is accompanied by its own view token $\mathbf{v}_{\text{src}}$, providing explicit source-target viewpoint correspondence for novel view synthesis.

each embedding with a specific camera pose, enabling viewpoint-controllable generation that is geometrically grounded in the shared 3D representation learned during pre-training.

Task Formulation. CT retains all four pre-training tasks and introduces three new view-conditioned tasks (Figure 5): Image $\rightarrow$ Posed Image, 3D $\rightarrow$ Posed Image, and Text $\rightarrow$ Posed Image. The new tasks share a common mechanism: a learnable view token $\mathbf{v}$ encoding the target camera extrinsics is prepended to the output image tokens in the generation stream. When the source is also an image, it is accompanied by its own view token $\mathbf{v}_{\text{src}}$ in the prefix, providing the model with explicit source-target viewpoint correspondence and enabling novel view synthesis. This design yields progressively richer 3D-aware capabilities—from single-image novel view synthesis (Image $\rightarrow$ Posed Image) to explicit 3D-controllable rendering (3D $\rightarrow$ Posed Image) to text-driven viewpoint-specific generation (Text $\rightarrow$ Posed Image).

Data. The view-conditioned tasks are constructed from the existing Image-3D pairs. For each 3D asset, we render images at $N = 6$ fixed canonical viewpoints (front, back, left, right, top, bottom), yielding posed image-3D tuples. During training, a task and a target viewpoint are randomly sampled per example. The pre-training data pools remain active to prevent catastrophic forgetting of the base generation capabilities.

Training Setup. CT uses the same training objective (Eq. 1) and optimizer configuration as pre-training. Only the N view token embeddings are newly initialized; all other parameters are warm-started from the pre-trained checkpoint. The learning rate is reduced to 1×10^{-5} with 1,000 warm-up steps. The model trains for 120,000 steps with an effective batch size of 1,024. The temperature-based data mixing strategy (Eq. 2) is extended to include the new view-conditioned pool, with priority weights adjusted to emphasize 3D-aware generation while retaining base capabilities.

4.6 Supervised Fine-Tuning (SFT)

Pre-training and Continued Training equip the model with cross-modal alignment and viewpoint awareness, but each task is still trained on pairwise data independently: there is no guarantee that a 3D mesh produced by Image $\rightarrow$ 3D renders back to the conditioning image, or that an image generated by Text $\rightarrow$ Image yields a plausible 3D reconstruction. The SFT stage closes this loop by introducing *interleaved multi-modal sequences* that chain image generation, 3D reconstruction,

Table 7 SFT task configurations. All tasks use text conditioning via the conditioning stream. $\mathbf{v}_k$ denotes view tokens. $\text{Img}_{\times N}$ denotes N posed images generated sequentially.

Task	Sequence
Text $\rightarrow$ Img $\rightarrow$ 3D	[Img ₁ , 3D]
Text $\rightarrow$ 3D $\rightarrow$ Img	[3D, $\mathbf{v}_1$, Img ₁]
Img $_{\times N} \rightarrow$ 3D	[$\mathbf{v}_1$, Img ₁ , ..., $\mathbf{v}_N$, Img _N , 3D]
3D $\rightarrow$ Img $_{\times N}$	[3D, $\mathbf{v}_1$, Img ₁ , ..., $\mathbf{v}_N$, Img _N]
Text $\rightarrow$ Img $_{\times N} \rightarrow$ 3D $\rightarrow$ Img $_{\times M}$	[$\mathbf{v}_1$, Img ₁ , ..., $\mathbf{v}_N$, Img _N , 3D, $\mathbf{v}_{N+1}$, Img _{N+1} , ...]

and novel view rendering into single autoregressive sequences. Because each subsequent modality is conditioned on all previously generated tokens, the training signal naturally enforces cross-modal consistency: a 3D mesh must be faithful to the images that precede it, and novel views rendered from that mesh must be coherent with the original generation.

Task Formulation. We define five SFT tasks (Table 7) covering all practical generation pipelines. Each task is formulated as a single autoregressive sequence over interleaved modality tokens. Text conditioning remains in the conditioning stream with the same CFG dropout ($p_{\text{drop}} = 0.1$). View tokens $\mathbf{v}_k$ specify camera poses for posed image generation.

5 Align to Instruction-Based 3D Editing

Earlier sections have demonstrated that Omni123’s unified autoregressive architecture excels not only at 3D generation but also at cross-modal understanding, owing to its inherited text-conditioned capabilities and the rich 3D–2D correspondences learned during pre-training and continued training. Consequently, we can adapt the model into an instruction-based 3D editing system, further revealing its potential to benefit users in practical content-creation workflows.

Task Formulation. We formulate instruction-based 3D editing as a conditional generation task as followed. Let $\mathcal{M}_{\text{src}}$ denote the source mesh tokenized into $L_m = 1,024$ discrete tokens $\mathbf{s}$ via Cube3D, and let c_{edit} be a natural-language edit instruction (e.g., “add wings to the car”, “change the material to gold”). Following the prefix-conditioned paradigm established in pre-training (Section 4.4), the source 3D shape tokens are placed as a non-causal prefix in the self-attention stream, while the edit instruction is injected via cross-attention. The model autoregressively generates the target token sequence $\mathbf{s}' = (s'_1, \dots, s'_{L_m})$, which is decoded into the edited mesh $\mathcal{M}_{\text{tgt}}$ via the Cube3D decoder and Marching Cubes. Text conditioning is randomly dropped with probability $p_{\text{drop}} = 0.1$ during training to enable Classifier-Free Guidance at inference, so the model is exposed to three conditioning regimes: edit instruction + source prefix, source prefix only, and unconditional, mirroring the CFG setup of the prefix-conditioned pre-training tasks.

Data. A key bottleneck for instruction-based 3D editing has been the lack of large-scale, high-quality paired data. Unlike 2D image editing, where paired datasets can be constructed at scale using foundation models [6], 3D editing data must additionally ensure cross-view consistency, structural fidelity in unedited regions, and semantic coherence of the edited parts. We adopt 3DEditVerse [77], the largest paired 3D editing benchmark to date, comprising 116,309 high-quality training pairs and 1,500 curated test pairs. Each pair consists of a source mesh, an edited mesh, and a natural-language edit instruction describing the modification. 3DEditVerse is constructed through two complementary automated pipelines: (i) a *pose-driven geometric* pipeline that leverages publicly available 3D characters and animation sequences, exploiting the fact that different poses of the same character naturally form valid “before–after” edit pairs for articulation and structural variations; and (ii) a *foundation-model-guided appearance* pipeline

that orchestrates multiple foundation models to generate edited assets with modified textures, colors, or stylistic attributes while preserving the underlying geometry. Both pipelines ensure edit locality, multi-view consistency, and semantic alignment between the instruction and the applied modification. Each training pair is tokenized into source tokens $\mathbf{s}$, target tokens $\mathbf{s}'$, and text embeddings via the same encoders used throughout Omni123.

Training Objective. The editing task shares the same autoregressive cross-entropy loss as all earlier training stages (Eq. 1), applied over the target token sequence:

$$\mathcal{L}_{\text{edit}} = - \sum_{k=1}^{L_m} \log p_{\theta}(s'_k \mid s'_{<k}, \mathbf{s}, c_{\text{edit}}) \quad (3)$$

where s'_k is the k -th target 3D token, $s'_{<k}$ denotes all preceding target tokens, $\mathbf{s}$ is the source prefix, and c_{edit} is the text conditioning (replaced with a learned null embedding when dropped). The loss is computed only over the target tokens; the source prefix is excluded from the loss computation. A uniform loss weight is applied across all target positions.

Training Setup. All parameters are initialized from the SFT checkpoint (Section 4.6). We fine-tune the full model using AdamW [48] ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay = 0.1) with a peak learning rate of 1×10^{-5} and a cosine schedule with 1,000 warm-up steps. Training proceeds for 10,000 steps with an effective batch size of 128 across 32×H100 GPUs. Gradient norms are clipped at 5.0, consistent with the pre-training configuration.

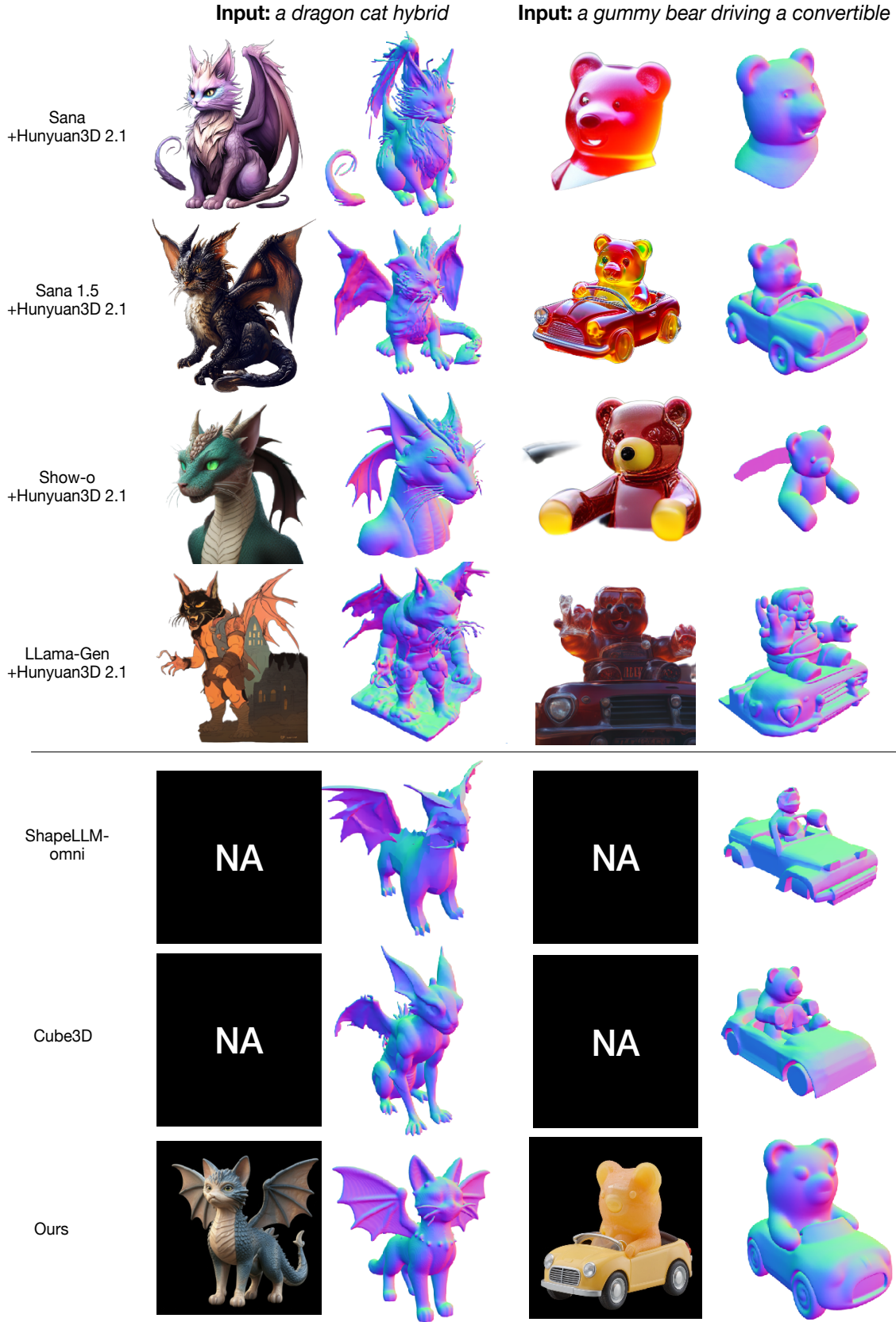


Figure 6 Qualitative comparison of text-to-3D shape generation across two compositional prompts. Rows 1–4: Two-stage Text→Image→3D methods, each showing the generated image (left) and reconstructed mesh as rendered image and normal map (right). Rows 5–6: Native text-to-3D baselines, which generate only meshes without intermediate images (marked “NA”). Row 7: Omni123 (Ours), which natively generates both a rendered image and a 3D mesh within a single autoregressive sequence.

6 Experiments

We evaluate Omni123 on text-to-3D shape generation (Section 6.1), instruction-based 3D editing (Section 6.2), and image tokenizer reconstruction quality (Section 6.3). Unless otherwise stated, all Omni123 results use greedy decoding with Classifier-Free Guidance scale $\omega = 5.0$.

6.1 Text-to-3D Shape Generation

As a cornerstone of 3D content generation, structurally consistent and high-quality shape generation provides the essential geometric scaffolding required for subsequent pipelines, including texturing, rendering, and physics simulation. Here, we evaluate the text-to-shape generation performance of Omni123, focusing on geometric fidelity and semantic alignment with input prompts.

Metrics. To evaluating semantic alignment, we uniformly sample 8,192 surface points from each generated mesh and compute point-text similarities using two established multi-modal models: ULIP [85](reporting ULIP-T) and Uni3d [96](reporting Uni3D-T). We evaluate on 100 diverse text prompts encompassing various object categories and styles, compositional descriptions and fine-grained attributes.

Baselines. To rigorously isolate the advantages of our **interleaved X-to-X training paradigm**, we benchmark against two dominant methodologies. *(i) Cascaded text-to-image-to-3D pipelines:* We pair leading image generators of *comparable parameter scales*—Sana-1.5 [81] (4.8B), Sana [80] (1.6B), Show-o [82] (1.3B), and LlamaGen [66] (0.775B)—with Hunyuan3D-2.1 [27]. These two-stage baselines represent indirect workflows that frequently compromise geometric consistency. *(ii) Native text-to-3D models:* ShapeLLM-Omni [88], a 7B autoregressive model that extends a pre-trained VLM (Qwen2.5-VL [2]) with 3D generation capabilities, and Cube3D [4], a 1.6B autoregressive model trained natively for text-to-3D generation. Both bypass the two-stage handoff but are constrained by 3D data scarcity and limited cross-modal transfer.

Quantitative Comparison. Table 8 demonstrates that two-stage pipelines consistently underperform native methods in **semantic-geometric alignment**, primarily bottlenecked simultaneously by the inherent prior gap between the image generator and the 3D lifter and inevitable error accumulation across the cascaded pipeline. Conversely, VLM-based native models avoid this handoff but are constrained by 3D data scarcity, in-effective knowledge transfer and sub-optimal modality fusion. By leveraging abundant 2D observations as a unified geometric prior, both Omni123 variants surpass these baselines by a clear margin. Notably, our **2B model** achieves superior alignment compared to the 7B ShapeLLM-Omni [88], demonstrating that interleaved cross-modal training is a fundamentally more *parameter-efficient* scalable route than blindly scaling isolated unimodal generators upon pretrained VLM backbone.

Qualitative Comparison. Figure 6 visually supports our quantitative findings on complex compositional prompts. Two-stage pipelines produce compelling intermediate images, but their image-to-3D handoff frequently collapses into artifact-ridden meshes—geometric flattening, disjointed limbs, and loss of fine-grained structure are common failure modes. Native 3D baselines bypass this fragile lifting step, yet their reliance on scarce 3D data limits compositional generalization: complex prompts often result in entangled features and semantically unfaithful geometry. Omni123 avoids both failure modes by generating a high-fidelity 2D image and a structurally sound mesh within a single autoregressive sequence, successfully articulating fine-grained anatomical features and precise multi-object spatial relationships.

6.2 Instruction-Based 3D Editing

As the most direct and natural interface for 3D creation, language instruction-driven 3D editing is a pivotal capability for broad 3D applications. We evaluate Omni123 on Edit3D-Bench [50], utilizing its curated source-target mesh pairs for geometric *addition* and *removal* operations. Our

Table 8 Quantitative comparison of text-to-3D shape generation. We evaluate shape-text alignment using ULIP-T [85] and Uni3D-T [96]. The two-stage methods pair various image generators with Hunyuan3D 2.1 [27] as the image-to-3D backbone. Best overall is **bolded**, second-best is underlined.

Method	Params	ULIP-T $\uparrow$	Uni3D-T $\uparrow$
<i>Two-stage: Text $\rightarrow$ Image $\rightarrow$ 3D (3D backbone: Hunyuan3D 2.1)</i>			
Sana-1.5 [81]	4.8B	0.1516	0.1672
Sana [80]	1.6B	0.1448	0.1589
Show-o [82]	1.3B	0.1447	0.1581
LlamaGen [66]	0.775B	0.0946	0.1067
<i>Native: Text $\rightarrow$ 3D</i>			
ShapeLLM-Omni [88]	7B	0.1649	0.2666
Cube3D [4]	1.6B	0.1726	0.2768
Ours	2.2B	0.1832	<u>0.2855</u>

Table 9 Quantitative comparison of image tokenizers on ImageNet-1K [16] 512×512 test set.

Method	FID $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$
<i>Continuous</i>			
Cosmos-CI16 $\times$ 16 [1]	0.356	28.60	0.7826
Flux.1 [37]	0.017	37.69	0.963
SDXL [54]	0.188	30.35	0.835
SVG-T2I [64]	1.994	20.60	0.587
Ours VAE	0.017	39.15	0.966
<i>Discrete</i>			
Cosmos-DI16 $\times$ 16 [1]	1.215	23.76	0.645
LlamaGen [66]	0.839	23.96	0.664
Show-o [82]	1.183	24.20	0.680
Ours VAE + 1D Tokenizer	0.432	24.61	0.759

goal is to perform semantically faithful and structurally articulated modifications to a source mesh closely following language instructions.

Metrics. To quantify editing fidelity, we report Chamfer Distance (CD) and F1 score (F1) between the generated and ground-truth edited meshes. Each test sample requires the model to resolve natural-language directives into precise, localized geometric transformations.

Baselines. We compare against four recent 3D editing methods: ShapeLLM-Omni [88], Steer3D [50], 3DEditFormer [77], and Tailor3D [56]. This selection strategically span both native autoregressive editors and optimization-based pipelines. While ShapeLLM-Omni features a native text-guided variant, its publicly available weights only support image-based inputs. To ensure maximum fairness and isolate each model’s intrinsic 3D editing and instruction-following capabilities, we intentionally employ the powerful Flux-Kontext [39] to provide high-quality 2D edited image input.

Quantitative Comparison. Table 10 highlights our superior geometric fidelity. Omni123 achieves the lowest Chamfer Distance (CD) across all tasks, indicating significantly tighter structural alignment with the ground-truth targets. While Steer3D leads in F1 score, we note that Edit3D-Bench is constructed using the same data pipeline as Steer3D’s training set,

Table 10 Comparison on add and remove editing tasks in Edit3D-Bench [50].

Method	Add		Remove		Avg	
	CD ↓	F1 ↑	CD ↓	F1 ↑	CD ↓	F1 ↑
Ours	0.0736	0.1743	0.0632	0.2259	0.0684	0.2001
ShapeLLM-Omni [88]	0.2546	0.0877	0.2237	0.1166	0.2392	0.1022
Steer3D [50]	0.1404	0.2414	0.0976	0.3044	0.1190	0.2729
3DEditFormer [77]	0.1676	0.1955	0.1342	0.1836	0.1509	0.1896
Tailor3D [56]	0.1661	0.1217	0.1755	0.1352	0.1708	0.1285

Table 11 Quantitative comparison of image tokenizers on ImageNet-1K [16] 512×512 test set.

Method	FID ↓	PSNR ↑	SSIM ↑
<i>Continuous</i>			
Cosmos-CI16 × 16 [1]	0.356	28.60	0.7826
Flux.1 [37]	0.017	37.69	0.963
SDXL [54]	0.188	30.35	0.835
SVG-T2I [64]	1.994	20.60	0.587
Ours VAE	0.017	39.15	0.966
<i>Discrete</i>			
Cosmos-DI16 × 16 [1]	1.215	23.76	0.645
LlamaGen [66]	0.839	23.96	0.664
Show-o [82]	1.183	24.20	0.680
Ours VAE + 1D Tokenizer	0.432	24.61	0.759

giving it a distributional advantage on this benchmark. Despite this, our consistent lead in CD demonstrates more precise execution of global geometric transformations and topological changes. These results suggest that the cross-modal representations learned through interleaved training translate effectively to editing tasks, enabling faithful instruction following even with relatively limited fine-tuning data (116K pairs).

6.3 Image Tokenizer Evaluation

As the representational foundation of our unified framework, a high-fidelity image tokenizer dictates essential structural and textural constraints and mediates the cross-modal information flow. We evaluate our two-stage tokenizer (Section 4.3) on the ImageNet-1K [16] 512×512 test set, benchmarking both the continuous VAE (Stage 1) and the full discrete 1D pipeline.

Metrics. We evaluate the tokenizer’s performance across three hierarchical dimensions: FID to measure the distributional alignment between reconstructed and ground-truth image manifolds, ensuring results are perceptually natural, realistic, and statistically sound; PSNR to quantify pixel-level fidelity; and SSIM to assess structural similarity and geometric integrity.

Quantitative Comparison. Table 11 highlights our superior reconstruction fidelity. In the continuous regime, our VAE surpasses the renowned Flux.1 in fidelity and structural consistency **using only ImageNet training set**. This underscores the exceptionally powerful prior inherited from pretrained representation encoder (DINOv3 [65]). Similarly, our discrete pipeline significantly outperforms recent state-of-the-art tokenizers by a large margin across all metrics. These results validate our two-stage strategy as an effective means of minimizing quantization

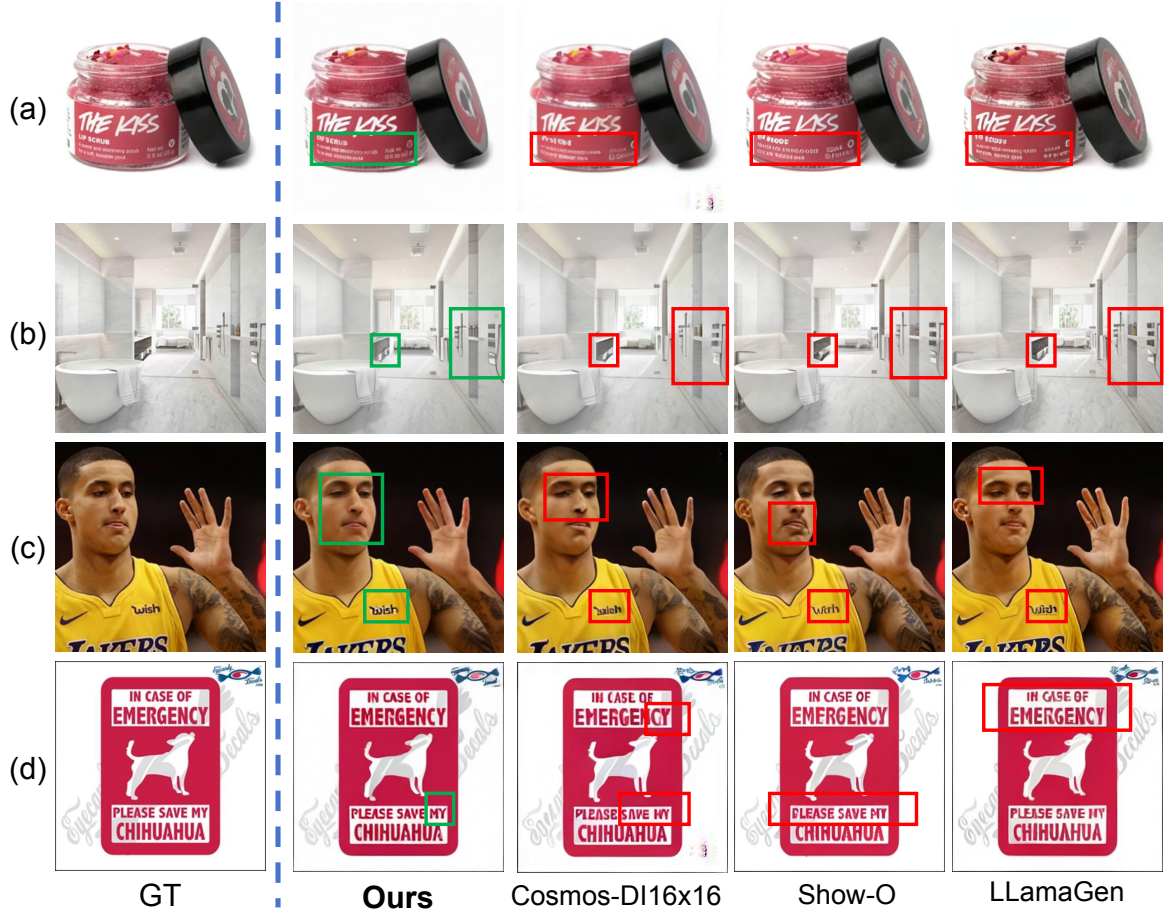


Figure 7 Qualitative comparison of discrete image tokenizers on web images of 512×512 resolution. **Our tokenizer** demonstrates superior reconstruction quality in (a) object-centric images, (b) scene-level details, (c) human details, and (d) text rendering.

degradation while ensuring a robust representational space.

Qualitative Comparison. Figure 7 illustrates the outstanding reconstruction quality achieved by our image tokenizer. While competing methods often suffer from significant blurriness and semantic distortion, Omni123 demonstrates a clear superiority in preserving high-frequency details and structural integrity. Notably, our model achieves **highly faithful text rendering and logo reconstruction** (Rows a, c, and d) which are particularly challenging for discrete tokenizers. Moreover, in **complex scenes** (Row b) and **human portraits** (Row c), our tokenizer preserves *sharp geometric boundaries* and *identity-consistent facial features*. These qualitative results are well aligned with the quantitative comparisons, further confirming that our **1D image tokenizer** effectively retains the geometric and semantic priors necessary to provide **high-fidelity guidance** for joint 3D synthesis.

7 Conclusion

We presented Omni123, a unified multimodal generative model that addresses 3D data scarcity by unifying text-to-2D and text-to-3D generation within a single autoregressive framework. Our three-stage training recipe—cross-modal pre-training over heterogeneous pairs, continued training with learnable view tokens for viewpoint awareness, and Text–Image–3D interleaved fine-tuning that closes the “semantic–visual–geometric” loop—enables abundant 2D observations to serve as an implicit structural constraint for 3D learning. Experiments on text-guided 3D generation and instruction-based 3D editing validate that this unified approach significantly improves geometric fidelity and semantic alignment, offering a scalable pathway toward multimodal 3D world models. Current limitations include the fixed mesh resolution of the 3D tokenizer and reliance on canonical viewpoints; future work will explore adaptive-resolution tokenization, scene-level generation, and material and physics modeling.

Acknowledgments. We thank Xiaohe Ma, Bing Yu, Jianfeng Xiang for insightful discussions and valuable suggestions. We thank the Meshy AI team for their contributions to infrastructure, data curation, and evaluation.



Figure 8 Additional qualitative showcase I. More examples of joint image-and-3D generation by Omni123.

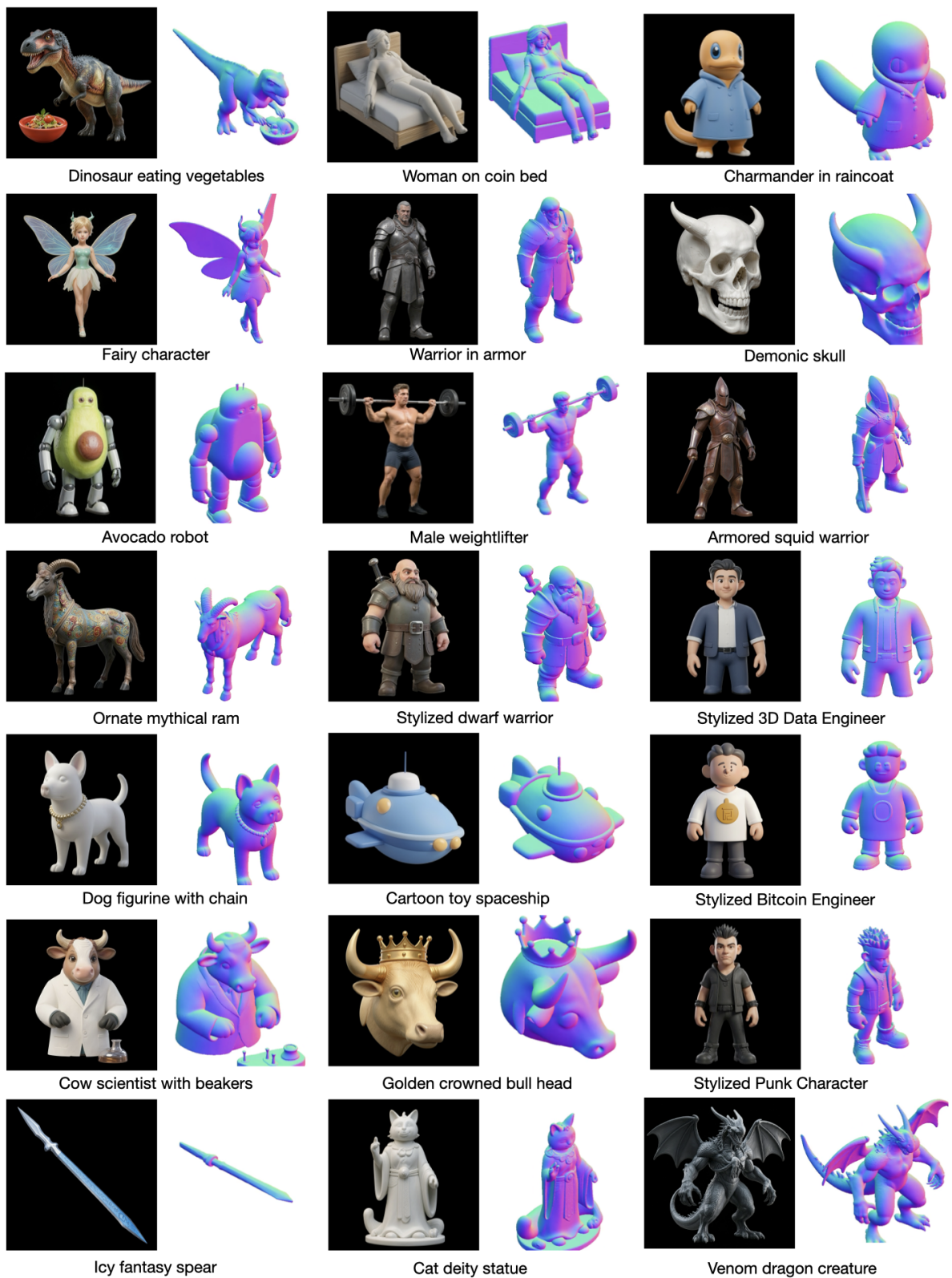


Figure 9 Additional qualitative showcase II. More examples of joint image-and-3D generation by Omni123.

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